

# **FreshCart Supply Chain Performance Analysis**

Prepared by: Saffan Nagani

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Tools Used: Microsoft Excel and Microsoft Word

Dataset: FreshCart Internal Order Data - 488 records, 9 suppliers,

5 warehouses, 5 product categories, January–December 2023

## Business Problem

FreshCart, a grocery supply chain operator, manages orders across 9 suppliers, 5 warehouses, and 5 product categories. Without a clear understanding of supplier delivery performance, warehouse distribution patterns, and the relationship between operational efficiency and profitability, inventory planning becomes reactive and costly.

Therefore, this analysis examines FreshCart's 2023 order data to identify performance gaps, quantify their business impact, and recommend targeted improvements across supplier management, warehouse operations, and inventory strategy.

## Data Cleaning Summary

The original dataset contained 500 records with multiple quality issues requiring cleaning before analysis. Key cleaning steps included:

- Removal of 12 duplicate Order IDs, leaving 488 records for analysis
- Standardized inconsistent date formats across Order\_Date and Delivery\_Date columns using TRIM, SUBSTITUTE, and DATEVALUE functions
- Cleaned inconsistent supplier name formatting affecting approximately 80 rows using TRIM and UPPER functions
- Flagged text-stored currency values in Unit\_Revenue column
- Identified and flagged 2-3 records with invalid Warehouse\_ID values recorded as 0 which are excluded from warehouse analysis
- Created Delivery\_Days column (Delivery\_Date minus Order\_Date) and the negative values are flagged as Data Errors
- Created On\_Time\_Flag column using promised lead time as benchmark, resulting in 433 On Time, 42 Late, and 13 Not Applicable records

## KPI Performance Summary

1. On-Time Delivery Rate: 88.7% - Target: 95% - Needs Improvement
2. Order Fill Rate: 99.4% - Target: 98% - Target Reached
3. Inventory Turnover: 2.03 - Target: 8-12 times - Below Target
4. Days of Stock: 179.7 days - Target: 30-45 days - Above Target

## Key Findings

### 1. NOVA COLD LOGISTICS is the weakest supplier

Nova Cold Logistics recorded 15 late deliveries out of 52 valid orders i.e., the worst on-time performance of any supplier. FRESH LINK PARTNERS also underperforms with 5 late deliveries. In contrast, BLUE STAR WHOLESALE delivered 69 out of 70 valid orders on time, making it FreshCart's most reliable supplier.

### 2. Frozen products are the highest risk category

Frozen products carry the highest average unit revenue at \$22.11 with a 35.2% profit margin, making them FreshCart's most valuable category per unit. However, Frozen also records the latest deliveries of any category (11), suggesting that supply chain failures are concentrated in the highest-value products. Every late Frozen delivery represents a missed opportunity on FreshCart's most profitable line.

### 3. Warehouse concentration creates single points of failure

WH-D handles 49% of all Bakery shipments (13,639 units) and WH-E handles 50% of all Produce shipments (15,060 units). This level of concentration means a disruption at either warehouse would critically impact supply of those categories with no immediate fallback.

### 4. Inventory turnover is critically below target

FreshCart's inventory turnover of 2.03 against a target of 8-12 times indicates that stock is moving extremely slowly through the supply chain. Days of Stock at 179.7 days means FreshCart is carrying almost 6 months of inventory which is far above the 30-45 day target. This ties up significant working capital and increases the risk of spoilage, particularly for perishable categories like Produce and Dairy.

## **5. Order fill rate is the only KPI meeting its target**

At 99.4% against a 98% target, FreshCart is fulfilling almost all customer orders completely. This is a genuine strength and should be maintained as delivery performance and inventory efficiency are improved.

## **Recommendations**

### **1. Place NOVA COLD LOGISTICS on a performance improvement plan**

With a 71% on-time delivery rate against a 95% target, Nova Cold Logistics is the single biggest driver of FreshCart's OTDR shortfall. Issue a formal performance review, set a 90-day improvement target of 90% OTDR, and identify backup suppliers for Frozen and Beverages categories where Nova Cold operates.

### **2. Prioritize Frozen category supplier reliability**

Given that Frozen products generate the highest revenue per unit and carry the latest deliveries, improving Frozen supply chain reliability should be the top priority. Negotiate stricter SLAs with Frozen suppliers and introduce penalty clauses for late deliveries exceeding the promised lead time.

### **3. Redistribute warehouse workload to reduce concentration risk**

WH-D and WH-E carry disproportionate share of Bakery and Produce respectively. Redistribute a minimum of 20% of WH-D Bakery volume to WH-B or WH-C and spread WH-E Produce volume across at least two additional warehouses to eliminate single points of failure.

### **4. Urgently address inventory turnover**

A turnover rate of 2.03 and 179.7 days of stock indicates a serious inventory management problem. Immediate actions should include a full stock audit to identify slow-moving and excess inventory, introduction of demand-driven replenishment to replace fixed ordering cycles, and category-specific turnover targets with weekly monitoring.

### **5. Maintain and monitor order fill rate**

The 99.4% fill rate is FreshCart's strongest operational metric. Ensure this is not compromised as inventory reduction initiatives are implemented. Fill rates should be tracked weekly alongside OTDR as the two primary supplier performance metrics.

## Conclusion

FreshCart's 2023 supply chain data reveal a business with strong order fulfillment capability but significant weaknesses in delivery reliability and inventory efficiency. The combination of a below-target OTDR (88.7%), critically low inventory turnover (2.03), and supplier concentration risk, particularly around Nova Cold Logistics and Frozen category deliveries. Addressing these three areas through targeted supplier management, warehouse redistribution, and demand-driven inventory planning would improve FreshCart's supply chain performance and protect its profit margins across all five product categories.